

NILAY KHISTY

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EDUCATION

- **Sardar Patel Institute of Technology** Nov 2022 - Jun 2026
Bachelor of Technology in Electronics and Telecommunications CGPA: 8.8
Minors in Computer Engineering
- **Indian Institute of Technology Madras** Jan 2023 - Dec 2025
Diploma in Data Science CGPA: 8.3
- **St Wilfred's School** Apr 2020 - May 2022
CBSE Class XII Percentage: 93%
- **DAV Public School** Apr 2018 - May 2020
CBSE Class X Percentage: 98%

PROJECTS

- **Fraud Detection Data Pipeline Project**
AWS S3, Snowflake - DBT, GitHub Actions, Power BI, Python, SQL
 - Processed 1.5M transactions via AWS S3 → Snowflake with dbt models reducing transformation time.
 - Implemented automated CI/CD using GitHub Actions, ensuring continuous testing and seamless promotion of models from development to production.
 - Delivered real-time fraud analytics dashboards in Power BI, enabling immediate visibility into transaction anomalies and risk patterns.
- **System Threat Forecaster**
Google Colab, SQL, Python, Jupyter Notebook
 - Engineered top-30 features using Random Forest on 73 raw features, boosting model efficiency and reducing dimensionality by 60%.
 - Trained stacked ensemble (Random Forest, XGBoost, LightGBM) with 300 estimators each and optimized max_depth and learning_rate for each base model.
 - Achieved 0.675 macro F1-score on validation set using Logistic Regression as a meta-learner, improving baseline by over 15%.
- **Numeric Recognition on FPGA**
Python, Matlab, Verilog
 - Deployed a fully connected ANN on FPGA for classifying 28×28 MNIST digits, achieving 87% accuracy on 10000 training samples.
 - Implemented forward and backpropagation, enabling on-chip learning without reliance on external or cloud compute.
 - Designed a fully isolated, self-contained system using fixed-point arithmetic, optimized for real-time inference within FPGA constraints.

TECHNICAL SKILLS

- **Programming Languages:** C++, Python
- **Database Systems:** MySQL, Snowflake
- **Specialized Area:** Data Science, Analytics, Machine Learning, Business Data Management
- **Python Libraries:** Numpy, Pandas, Scikitlearn, Scipy, Matplotlib, Keras, Seaborn
- **Course work:** Data Structures, DBMS, Computer Networks, Operating Systems, OOPS

EXTRACURRICULAR

Training and Placement Office *Statistics Team*

- Managed and updated student placement data with across branches and recruiters.
- Analyzed trends to highlight placement growth and support institute reporting.